

3.2 Hydrocarbons

Question Paper

Course	CIEA Level Chemistry
Section	3. Organic Chemistry
Topic	3.2 Hydrocarbons
Difficulty	Medium

Time allowed: 70

Score: /58

Percentage: /100

Question 1a

A reaction scheme involving cyclohexane is shown in Fig. 4.1.

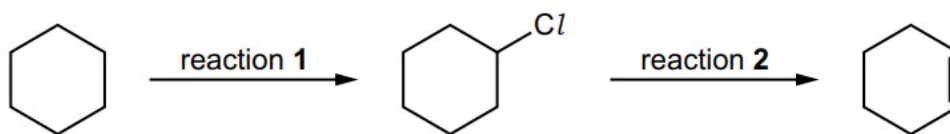


Fig. 4.1.

Reaction 1 involves a free radical substitution mechanism.

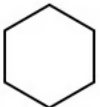
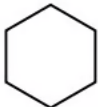
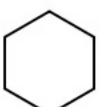
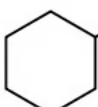
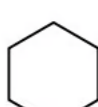
State the essential condition required for reaction 1 to occur.

[1 mark]

Question 1b

Complete Table 4.1 to give details of the mechanism in reaction 1. Include curly arrows to show the movement of electrons occurring in the termination step.

Table 4.1

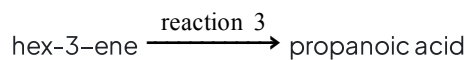
name of step	equation
.....	$\text{Cl}_2 \longrightarrow 2\text{Cl}\cdot$
propagation	 $+ \text{Cl}\cdot \longrightarrow$  $+ \dots\dots\dots$
.....	 $+ \text{Cl}_2 \longrightarrow$  $+ \text{Cl}\cdot$
termination	 $\longrightarrow$ 

[5 marks]**Question 1c**

Deduce the type of reaction that occurs in reaction 2.

[1 mark]**Question 1d**

Hex-3-ene is an isomer of cyclohexane. Hex-3-ene can be converted into propanoic acid.



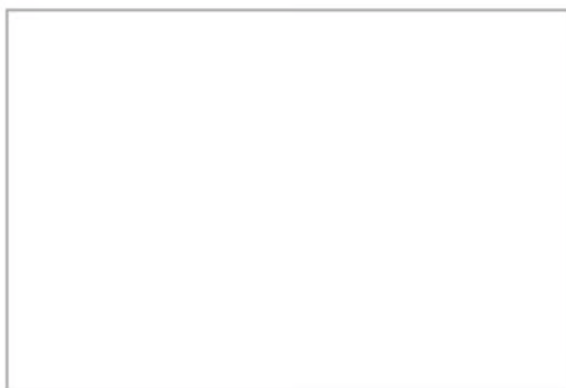
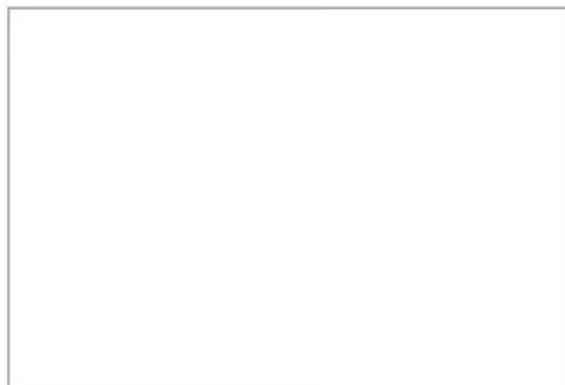
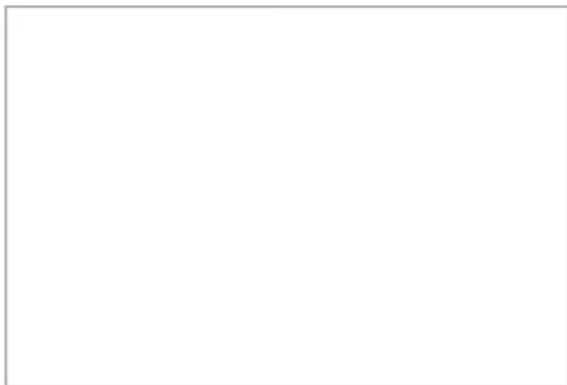
Deduce the reagents and conditions for reaction 3.

[2 marks]

Question 2a

This question is about alkenes with the molecular formula C_5H_{10} .

Draw the **skeletal** formulae of three **branched** chain alkenes with the molecular formula C_5H_{10} .



[3 marks]

Question 2b

An alcohol can react with concentrated phosphoric acid, H_3PO_4 , to form pent-2-ene.

i)

Give the formula of the alcohol that will form only pent-2-ene during this reaction.

[1]

ii)

Give a name for this type of reaction.

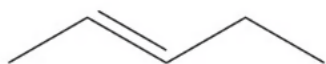
[1]

[2 marks]

Question 2c

Pent-2-ene reacts with hydrogen bromide, HBr , to form two bromoalkanes.

Complete the diagram to show the mechanism for the formation of 2-bromopentane in this reaction. Include curly arrows, and relevant lone pairs and dipoles.



[4 marks]

Question 2d

A sample of pent-1-ene, with a mass of 1.33 g, is warmed to 60°C in a sealed container. The volume of the container is 500 cm³.

Calculate the pressure inside the container.

Include units and give your answer to an appropriate number of significant figures.

[molar gas constant (R) = 8.31 J mol⁻¹ K⁻¹]

[4 marks]

Question 3a

Crude oil contains a mixture of hydrocarbons which are separated using fractional distillation.

Further processes can then be carried out, including cracking.

i)

What is meant by the term 'cracking'.

[1]

ii)

Explain why cracking is carried out.

[1]

[2 marks]

Question 3b

i)

State **two** conditions required for the cracking of alkanes.

[2]

ii)

Construct a balanced equation for the formation of heptane, C_7H_{16} and two other products, by cracking tetradecane, $C_{14}H_{30}$.

[1]

[3 marks]

Question 3c

Another alkane, C_8H_{18} is used in gasoline. It undergoes combustion to produce a mixture of products including carbon monoxide, oxides of nitrogen and unburnt hydrocarbons.

i)

Explain why carbon monoxide is dangerous if inhaled.

[2]

ii)

Describe the environmental effect of oxides of nitrogen and unburnt hydrocarbons.

[2]

[4 marks]

Question 3d

Catalytic converters are found in the exhaust system of cars.

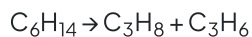
Use equations to show how a catalytic converter reduces the levels of carbon monoxide and oxides of nitrogen released into the atmosphere.

[2 marks]

Question 4a

This question is about alkanes and alkenes.

Propene and propane can be produced from the cracking of hexane, C_6H_{14} .



Explain why propane is relatively unreactive.

[4 marks]

Question 4b

Describe and give the result of a test that can be done to distinguish between the two hydrocarbons.

[2 marks]

Question 4c

Propene, C_3H_6 , is an alkene.

The reagents needed for three reactions of propene are shown in Fig. 4.1.

Complete Fig. 4.1 by drawing the structure of the organic product produced in each reaction.

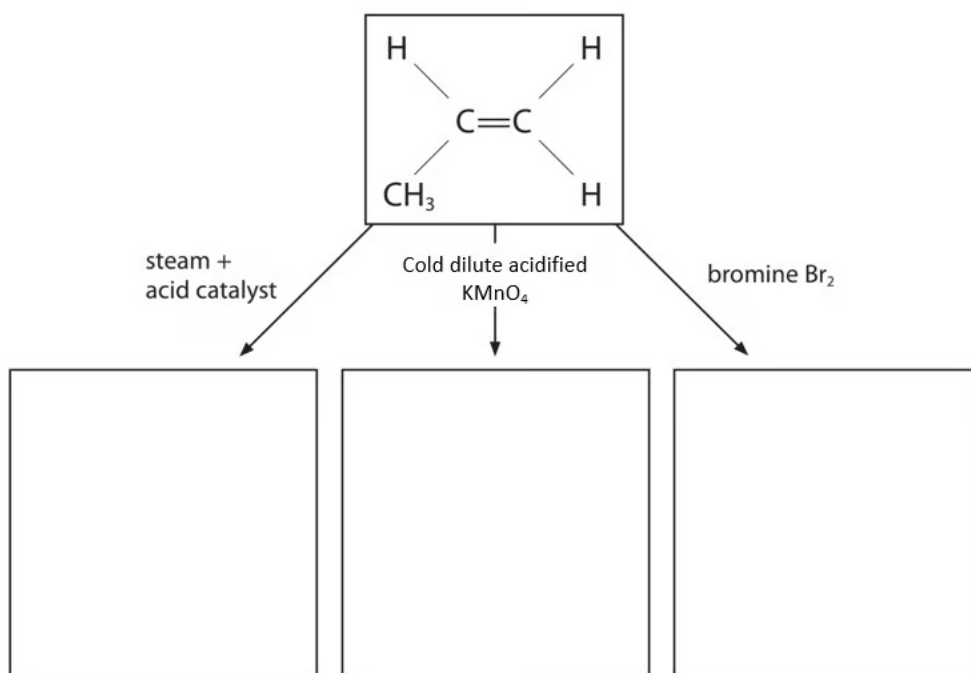


Fig. 4.1

[3]

ii)

Propene also reacts with hydrogen bromide, HBr.

Give the mechanism for this reaction to form the **major** product.

Include curly arrows, and relevant lone pairs and dipoles.

[3]

[6 marks]**Question 4d**

Propene reacts by addition polymerisation to form poly(propene).

Draw the structure of poly(propene), showing **two** repeat units.

[2 marks]

Question 5a

This question is about alkenes.

One method of preparing ethene in a school or college laboratory is from ethanol by using the apparatus shown below in Fig. 5.1.

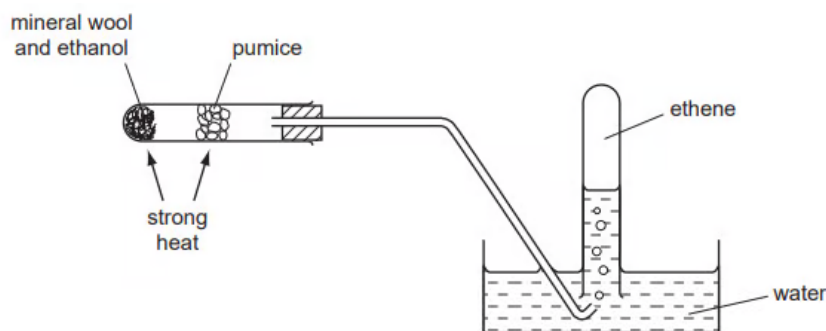


Fig. 5.1

i)
Write a balanced equation for this reaction.

[1]

ii)
What type of reaction is this?

[1]

[2 marks]

Question 5b

Ethene is bubbled into two separate test-tubes, one containing aqueous hydrogen bromide and the other containing cold, dilute acidified potassium manganate(VII).

In each case, complete Table 5.1 to describe any colour changes you would see and give the structural formula of the organic product.

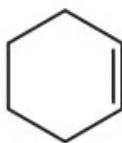
Table 5.1

	aqueous hydrogen bromide	cold, dilute acidified potassium manganate(VII)
colour at start		
colour after reaction		
structural formula of organic product		

[4 marks]

Question 5c

Cyclohexene has the following structural formula.



What is the molecular formula of cyclohexene?

[1 mark]

Question 5d

Cyclohexene can react with bromine.

i)
Draw the structural formula of the compound formed when cyclohexene is reacted with bromine.

[1]

ii)
State what type of mechanism is involved in forming this product.

[2]

[3 marks]

Question 5e

Draw the structural formula of the compound formed when cyclohexene is reacted with hot concentrated acidified potassium manganate(VII).

[1 mark]

Model Answers: Medium

1a

a) The essential condition required for reaction 1 to occur is:

- Ultraviolet / UV light; [1 mark]

[Total: 1 mark]

- You should recognise the first reactant as the alkane, cyclohexane and recognise that during reaction 1 one of its hydrogen atoms is substituted by a chlorine atom – this occurs by free radical substitution
- You should know that the condition required for this substitution reaction is UV light
- Alkanes are very unreactive so the reaction requires a highly reactive chlorine free radical, formed when a chlorine molecule is broken by UV light

1b

b) The completed table is:

name of step	equation
initiation	$\text{Cl}_2 \longrightarrow 2\text{Cl}\cdot$
propagation	
propagation	
termination	

- Initiation; [1 mark]
- HCl; [1 mark]
- Propagation; [1 mark]
- For the termination step: Correct species; [1 mark]
- Curly arrows; [1 mark]

[Total: 5 marks]

- Free radical substitution is a mechanism frequently tested
- Ensure you know the names of the different stages and the reactions that occur
- **Careful:** During the termination stage, an electron from 2 free radicals are shared to form a new bond, the movement of a single electron is shown by a single-headed arrow

1c

c) The type of reaction that occurs in reaction 2 is:

- Elimination; [1 mark]

[Total: 2 marks]

- During reaction **2**, the halogenoalkane forms an alkene
- In this reaction, HCl is eliminated:
 - $\text{C}_6\text{H}_{11}\text{Cl} \rightarrow \text{C}_6\text{H}_{10} + \text{HCl}$
- The conditions required for this reaction are heat and ethanolic sodium hydroxide
- Halogenoalkanes can undergo nucleophilic substitution to form an alcohol if NaOH (aq) is used

1d

d) The reagents and conditions for reaction **3** are:

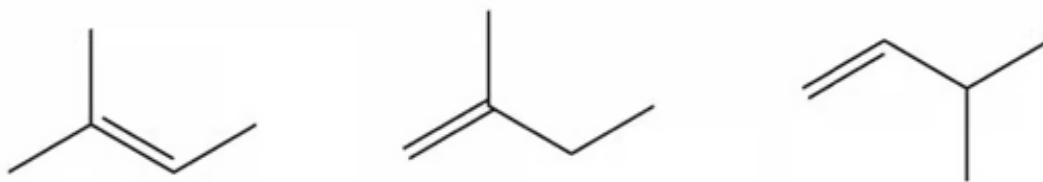
- **Reagents:** acid **AND** potassium manganate(VII); [1 mark]
- **Conditions:** hot **AND** concentrated; [1 mark]

[Total: 2 marks]

- Reaction **3** involves the reaction of an alkene to form a carboxylic acid
- This happens in an oxidation reaction, however, careful with stating the reagents and conditions as they determine the product formed
- With hot concentrated acidic potassium manganate(VII) as the oxidising agent, the double bond in hex-3-ene will be broken, forming propanal initially, which is then further oxidised to form propanoic acid
- However, with cold dilute KMnO_4 , the double bond opens up to form a diol, in this case, hexan-3,4-diol
- You need to know the different products formed using these different conditions and reagents

2a

a) The **skeletal** formulae of three **branched** chain alkenes with the molecular formula C_5H_{10} are:



- Each correct skeletal formula; [1 mark]
- Three correct **displayed** or **structural** formulae; [2 marks]
- For any two correct **displayed** or **structural** formulae; [1 mark]

[Total: 3 marks]

- **Careful:** You are asked for the **branched** chain alkenes of C_5H_{10} and **skeletal** formulae
 - Therefore the straight chain isomer is not awarded a mark

2b

b)

i) The formula of the alcohol that will form only pent-2-ene during this reaction is:

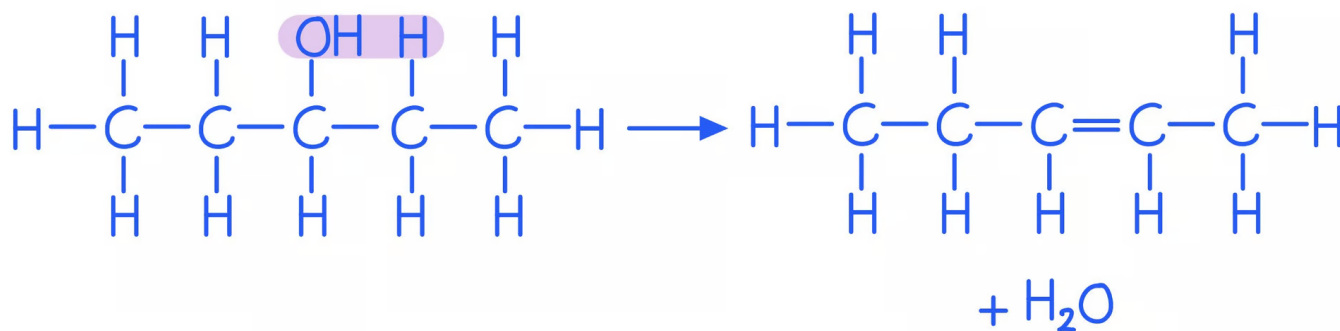
- $CH_3CH_2CH(OH)CH_2CH_3$; [1 mark]

ii) This type of reaction is:

- Dehydration
OR
Elimination; [1 mark]

[Total: 1 mark]

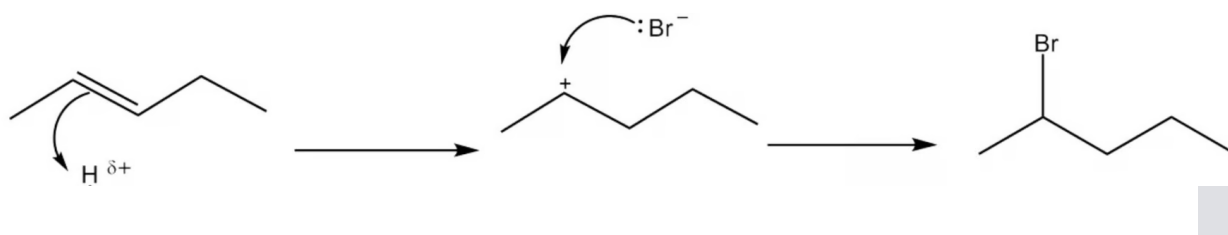
- When an alcohol reacts with concentrated phosphoric acid, the $-OH$ group and a hydrogen atom are lost from the alcohol to form an alkene and water
- The loss of water labels this reaction as dehydration as well as elimination
- Pentan-2-ol, $CH_3CH_2CH(OH)CH_2CH_3$ can **only** form pent-2-ene, which can be written as $CH_3CHCHCH_2CH_3$ or $CH_3CH_2CHCHCH_3$



- You will not score the mark for:
 - Giving the formula of pentan-1-ol as this will form pent-1-ene
 - Giving the formula of pentan-2-ol as this could form either pent-1-ene or pent-2-ene and the question specifically asks

2c

c) The completed diagram to show the mechanism for the formation of 2-bromopentane in this reaction is:

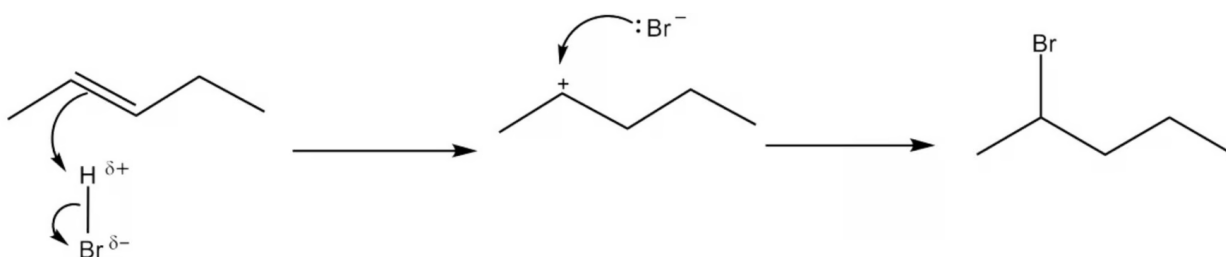




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2c

c) The completed diagram to show the mechanism for the formation of 2-bromopentane in this reaction is:



- Arrow from double bond to δ^+ H in HBr; [1 mark]
- Arrow from bond in HBr to δ^- Br; [1 mark]
- Structure of carbocation; [1 mark]
- Arrow from lone pair on Br^- to C^+ in carbocation and final products; [1 mark]

[Total: 4 marks]

- You must show the dipole to gain full marks, but you will only be penalised once for missing it out
- Your arrows must also be going in the right direction, from an area of negative charge to an area of positive charge
- When drawing a mechanism make sure you have the correct arrows
 - A full headed arrow $\rightarrow$ shows the movement of a pair of electrons

2d

d) To calculate the pressure inside the container:

- Calculation of moles of pent-1-ene:**

$$\text{Moles of pent-1-ene} = \frac{1.33}{70} = 0.019 \text{ (mol)}; [1 \text{ mark}]$$

- **Conversion of volume and temperature:**

$$\text{Volume} = 500 \times 10^{-6} \text{ m}^3$$

AND

$$\text{Temperature} = 333 \text{ K}; [1 \text{ mark}]$$

- **Rearrangement of ideal gas equation and calculation of p :**

$$p = \frac{(nRT)}{V} = \frac{(0.019 \times 8.31 \times 333)}{500 \times 10^{-6}} = 105154.74; [1 \text{ mark}]$$

- **Final answer to 2 or 3 SF and units:**

$$p = 105000 \text{ Pa} / 1.05 \times 10^5 \text{ Pa} / 1.1 \times 10^5 \text{ Pa}; [1 \text{ mark}]$$

[Total: 4 marks]

- The correct answer with units and no working scores 4 marks, though it is always best to show your working
- You can use N m^{-2} for Pa and 105 kPa

3a

a)

i) The term cracking means:

- Breaking large hydrocarbons into smaller hydrocarbons; [1 mark]

ii) Cracking is carried out because:

- Smaller hydrocarbons are more useful

OR

Smaller hydrocarbons are more in demand; [1 mark]

[Total: 2 marks]

- Cracking produces smaller alkanes, which are useful as fuels, and alkenes which are useful starting compounds to make other materials

3b

b)

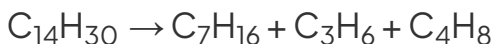
i) **Two** conditions required for the cracking of alkanes are:

- High temperatures; [1 mark]
- Aluminium oxide / Al_2O_3 catalyst; [1 mark]

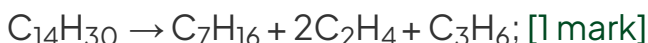
ii) A balanced equation for the formation of heptane, C_7H_{16} , by cracking tetradecane, $\text{C}_{14}\text{H}_{30}$:



OR



OR



[Total: 3 marks]

- Make sure you are specific for part (i)
 - You will not score the mark unless you specifically give aluminium oxide as a catalyst
 - You will also not score the mark if you state 'heat'
 - Cracking is carried out at approximately 550°C so heat is an inadequate word to describe this condition
- Write and balance an equation for cracking as you would any other equation
- The number of carbon and hydrogen atoms in the reactants and products should be equal
- **Remember:** A mixture of alkanes and alkenes is formed
- You will not score the mark for $\text{C}_{14}\text{H}_{30} \rightarrow \text{C}_7\text{H}_{16} + \text{C}_7\text{H}_{14}$ as only one other product has been formed

3c

c)

i) Carbon monoxide is dangerous if inhaled because:

- The carbon monoxide / CO binds well to haemoglobin which therefore cannot bind oxygen or carbon dioxide; [1 mark]
- Oxygen can not be delivered to organs

OR

Carbon dioxide can not be removed from the body; [1 mark]

ii) Two environmental effects of oxides of nitrogen and unburnt hydrocarbons are:

Oxides of nitrogen

Any **one** from:

- Acid rain; [1 mark]
- Limestone corrosion; [1 mark]
- Killing of aquatic life; [1 mark]
- Respiratory effects; [1 mark]

Unburnt hydrocarbons

- Smog / Global dimming; [1 mark]

[Total: 4 marks]

- CO is a toxic and odourless gas which can cause dizziness, loss of consciousness and eventually death
- We often talk about carbon monoxide inhibiting the transport of oxygen around the body, but it also prevents the removal of the waste gas carbon dioxide
- Part (ii) is worth two marks so one effect for each pollutant is required
- Unburnt hydrocarbons are also referred to as VOCs

3d

d) Catalytic converters reduce the levels of carbon monoxide and oxides of nitrogen released into the atmosphere because:

- $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$; [1 mark]
- $2\text{CO} + 2\text{NO} \rightarrow 2\text{CO}_2 + \text{N}_2$; [1 mark]

[Total: 2 marks]

- **Remember:** In a catalytic converter, the carbon monoxide is converted to carbon dioxide and oxides of nitrogen are converted to nitrogen
- The questions says to use equations so if no equations are present you will not score each mark

4a

a)

i) Propane is relatively unreactive because:

- Single covalent bonds need lots of energy to break; [1 mark]
- Propane is non-polar; [1 mark]

[Total: 2 marks]

- There is very little difference between the electronegativities of carbon and hydrogen, so there are no partial positive or negative charges and are non-polar
- They therefore don't react with polar reagents

4b

b) Describe and give the result of a test that can be done to distinguish between the two hydrocarbons:

- Add bromine water; [1 mark]
- Propene will decolourise / turn colourless

OR

Turns from orange to colourless; [1 mark]

[Total: 2 marks]

- Bromine water is a common test for unsaturation, the presence of a double bond

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4a

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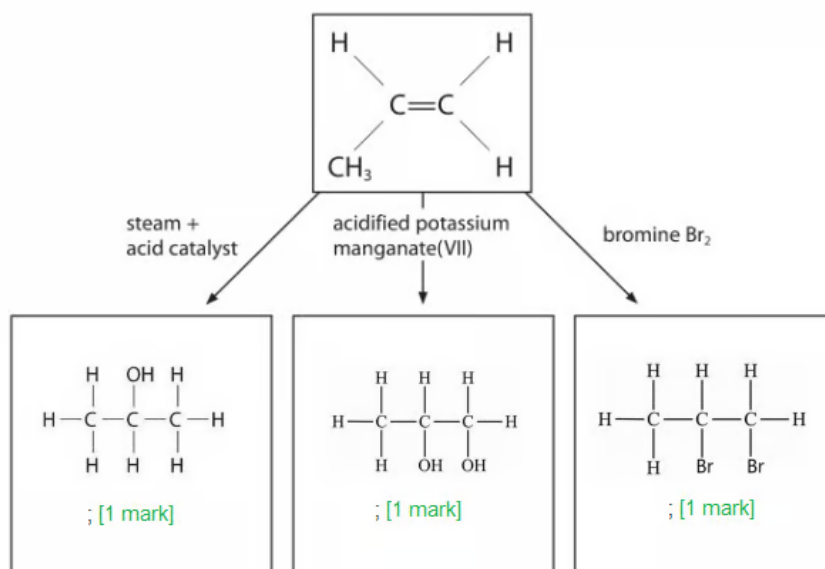
[Total: 2 marks]

- Bromine water is a common test for unsaturation, the presence of a double bond
 - Bromine is an orange solution
 - When bromine adds across the double bond in propene to form 1,2-dibromopropane, the solution decolourises
 - The lack of double bond in propane means the solution will remain orange

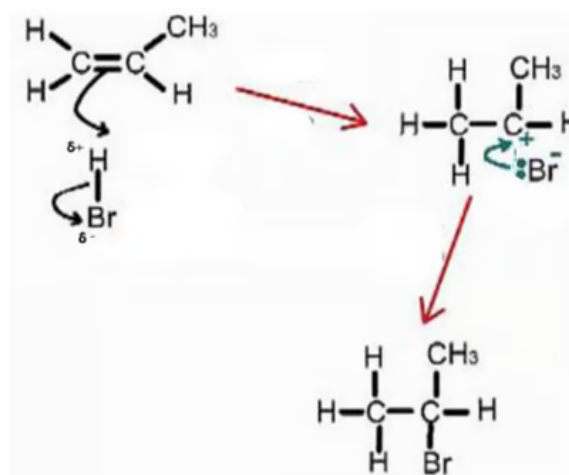
4c

c)

i) The organic product for each reaction is:



ii) The mechanism for this reaction to form the **major** product is:



- Dipole on HBr and two correct curly arrows; [1 mark]
- Correct intermediate; [1 mark]
- Curly arrow from lone pair on Br^- to C^+ or the space between the Br^- to C^+ ; [1 mark]

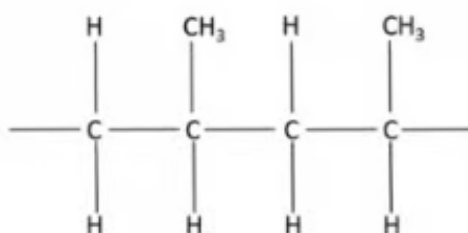
[Total: 6 marks]

- As you can see, you must know the reactions of alkenes in detail:
 - Alkene and steam react to form an alcohol
 - Alkene and hydrogen react to form an alkane

- Alkene and halogens react to form halogenoalkanes
- Alkenes are oxidised by cold dilute acidified potassium manganate (VII) to form a diol
- The curly arrows in part ii) must start from the covalent bond
 - From the H—Br bond it must go to the Br or beyond
 - From the C=C bond it must go to the H or in the space
- If, by mistake, you have drawn the mechanism between propene and Br₂ you can still score marks 2 and 3
- The major product is 2-bromopropane as the secondary carbocation is more stable than the primary carbocation
- If you have drawn a mechanism to show the formation of the minor product, 1-bromopropane then you cannot score marking point 2 for the intermediate

4d

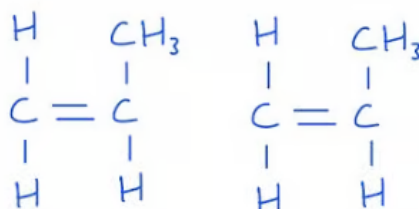
d) The structure of poly(propene), showing **two** repeat units is:



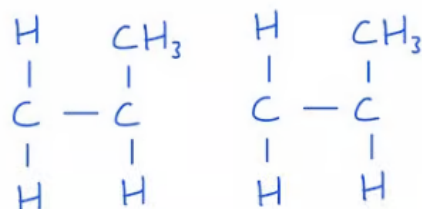
- Poly(propene) structure containing two repeat units with extension bonds; [1 mark]

[Total: 1 mark]

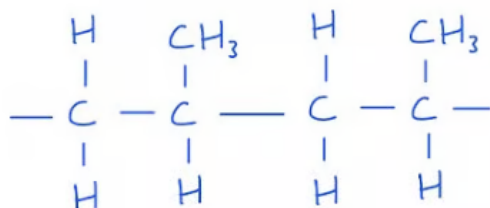
- To draw the structure with two repeating units, first, draw two monomers:



- Remove the C=C double bond, leaving a C-C single bond:



- Add a bond between the two carbon atoms adjacent to each other and add continuation bonds:



5a

a)

i) A balanced equation for this reaction is:

- $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$; [1 mark]

ii) The type of reaction is:

- Elimination

OR

Dehydration; [1 mark]

[Total: 2 marks]

- You should know the formulae for ethanol and ethene and be able to put them into an equation
- It is known as elimination as a molecule is eliminated from the structure
- It is also a dehydration reaction due to the loss of water
- In this example, pumice has been used as a catalyst but you should also be aware that aluminium oxide or a concentrated acid can be used on an industrial scale

5b

b) The completed table is:

	aqueous hydrogen bromide	cold, dilute acidified potassium manganate(VII)
colour at start	colourless	purple OR pink
colour after reaction	colourless	colourless or decolourised
structural formula of organic product	$\text{CH}_3\text{CH}_2\text{Br}$	$\text{HOCH}_2\text{CH}_2\text{OH}$

With hydrogen bromide:

- From colourless to colourless; [1 mark]
- $\text{CH}_3\text{CH}_2\text{Br}$; [1 mark]

With potassium manganate(VII)

- From purple / pink to colourless / decolourised; [1 mark]
- $\text{HOCH}_2\text{CH}_2\text{OH}$; [1 mark]

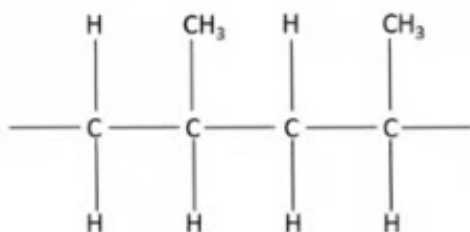
[Total: 4 marks]

- **Careful:** Say clear instead of colourless and you will not score the mark
- Both colours must be correct to achieve the first marking point for each chemical

- Alkenes are oxidised by cold dilute acidified potassium manganate (VII) to form a diol
- The curly arrows in part ii) must start from the covalent bond
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4d

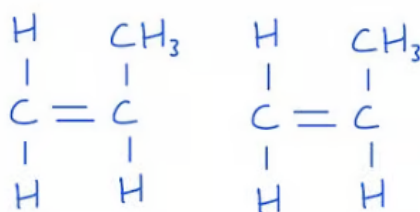
d) The structure of poly(propene), showing **two** repeat units is:



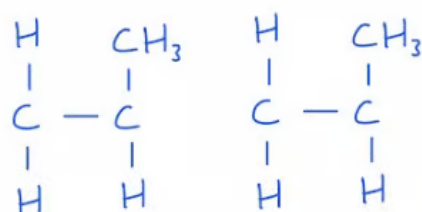
- Poly(propene) structure containing two repeat units with extension bonds; [1 mark]

[Total: 1 mark]

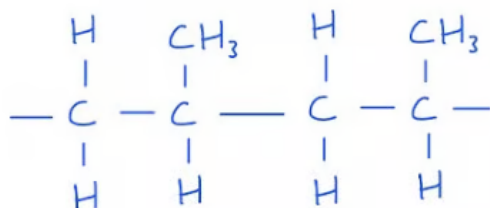
- To draw the structure with two repeating units, first, draw two monomers:



- Remove the C=C double bond, leaving a C-C single bond:



- Add a bond between the two carbon atoms adjacent to each other and add continuation bonds:



5a

a)

i) A balanced equation for this reaction is:

- $\text{C}_2\text{H}_5\text{OH} \rightarrow \text{C}_2\text{H}_4 + \text{H}_2\text{O}$; [1 mark]

ii) The type of reaction is:

- Elimination

OR

Dehydration; [1 mark]

[Total: 2 marks]

- You should know the formulae for ethanol and ethene and be able to put them into an equation
- It is known as elimination as a molecule is eliminated from the structure
- It is also a dehydration reaction due to the loss of water
- In this example, pumice has been used a catalyst but you should also be aware that aluminium oxide or a concentrated acid can be used on an industrial scale

5b

b) The completed table is:

	aqueous hydrogen bromide	cold, dilute acidified potassium manganate(VII)
colour at start	colourless	purple OR pink
colour after reaction	colourless	colourless or decolourised
structural formula of organic product	$\text{CH}_3\text{CH}_2\text{Br}$	$\text{HOCH}_2\text{CH}_2\text{OH}$

With hydrogen bromide:

- From colourless to colourless; [1 mark]
- $\text{CH}_3\text{CH}_2\text{Br}$; [1 mark]

With potassium manganate(VII)

- From purple / pink to colourless / decolourised; [1 mark]
- $\text{HOCH}_2\text{CH}_2\text{OH}$; [1 mark]

[Total: 4 marks]

- **Careful:** Say clear instead of colourless and you will not score the mark
- Both colours must be correct to achieve the first marking point for each chemical
- When an alkene reacts with a hydrogen halide, a halogenoalkane is formed
- When an alkene reacts with cold, dilute acidified potassium manganate(VII), it is oxidised to form a diol
 - Alkenes also react with hot concentrated potassium manganate(VII) so make sure you learn the difference between the two reactions

5c

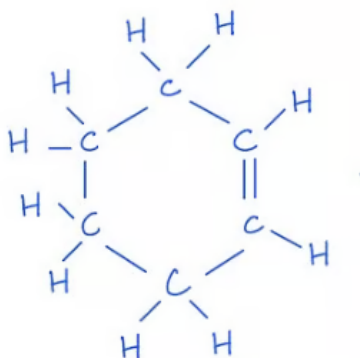
c) The molecular formula of cyclohexene is:

- C_6H_{10} ; [1 mark]

[Total: 1 mark]

- The molecular formula is the actual number of atoms of each element in a compound
- It can be useful to draw the displayed formula for the substance so that you can see the number of hydrogen atoms clearly

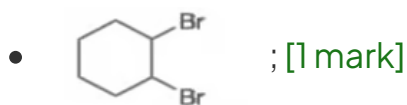
- **Remember:** Each carbon forms four covalent bonds



5d

d)

i) The structural formula of the compound formed when cyclohexene is reacted with bromine is:



ii) The type of mechanism is involved in forming this product is:

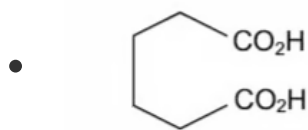
- Electrophilic; [1 mark]
- Addition; [1 mark]

[Total: 3 marks]

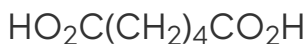
- Don't be put off with the question asking about a cycloalkene
- Bromine will add across the double bond, in exactly the same way it would if it was a straight chained alkene
- Make sure the bromine atoms are joined to the carbon atoms that had the double bond
- **Remember:** It is **electrophilic** addition as the double bond is susceptible to attack from **electrophiles**, in this case the bromine
- Bromine is **added** onto the structure hence it is an **addition** reaction

5e

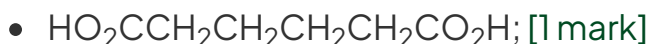
e) The structural formula of the compound formed when cyclohexene is reacted with hot concentrated acidified potassium manganate(VII) is:



OR



OR



[Total: 1 mark]

- When hot concentrated KMnO_4 is added, the carbon carbon double bond completely breaks and the diol initially formed can become oxidised further
- In cyclohexene, each carbon atom in the double bond is bonded to one H atom and one alkyl group which result in the formation of a carboxylic acid
- You can write the functional group out fully $-\text{COOH}$ to also score the marks
- The product formed will depend on the groups attached to the carbon atoms in the double bond